

Web Dev + Data ➡ AI Engineer

A focused, build-first plan that turns your **software & data** foundation into a hireable **GenAI / LLM application engineer** profile in six months.

WHERE YOU ARE Current Position

Working Professional · Web Development + Data Analysis. You already ship software and speak data — a rare, high-leverage base that puts you ahead of most aspiring AI engineers.

- Python
- APIs & deploy
- SQL / pandas
- Statistics
- Frontend

WHERE YOU'RE GOING Target Goal

AI Engineer — GenAI / LLM Applications. Build production AI products on top of foundation models. It's the fastest, most hireable lane for your profile — your web-dev edge compounds here.

- RAG
- Agents
- LLM APIs
- Evaluation
- MLOps-lite

HONEST DELTA Skill Gap Analysis

SKILL AREA	YOU HAVE	TARGET LEVEL	GAP
Engineering & data — Python, APIs, SQL	Strong	Maintain	Low
ML fundamentals	Light	Working	Medium
Deep learning — PyTorch, transformers	None	Conversational	Medium
LLM core — prompting, embeddings, RAG	None	Expert	High
Agents & orchestration — tools, MCP	None	Strong	High
Vector DBs & retrieval	None	Working	Medium
Evaluation, MLOps & serving	Partial	Working	Medium

ORDERED PRIORITIES Recommended Learning Plan

- 1

ML foundations

Andrew Ng ML Spec / fast.ai

Wk 1-3
- 2

Deep learning

PyTorch, neural nets, transformer intuition

Wk 4-6
- 3

LLM core

Prompting, embeddings, RAG, vector DBs

Wk 7-10
- 4

Agents

Function calling, LangChain / LlamaIndex, MCP

Wk 11-13
- 5

Eval & deploy

LangSmith, FastAPI, Docker, cloud

Wk 14-16
- 6

Polish & interview prep

Portfolio, system design, AI eng patterns

Wk 17-24

PROOF OF SKILL Suggested Projects

- > **RAG Knowledge Assistant** — embeddings + vector DB + FastAPI, wrapped in your own web UI.
- > **Multi-Tool AI Agent** — function calling / MCP that automates a real workflow end-to-end.
- > **Fine-Tuned Mini-Model** — LoRA on a niche task, shipped with an evaluation harness.
- > **Full-Stack AI SaaS** — deployed live; showcases the web-dev edge most candidates lack.

- Ship public
- Write it up
- Deploy live

GET SEEN, GET REFERRED Networking Strategy

- > **Build in public:** 3 LinkedIn posts/week documenting each project & lesson.
- > **Ship to GitHub** + publish 1 technical blog per project.
- > **Contribute** to 1-2 open-source GenAI repos.
- > **5 warm chats/month** with AI engineers & hiring managers; join one AI community.

MONTH BY MONTH Monthly Milestones

- M1

FOUNDATIONS

ML + DL basics; dev env live; LinkedIn revived.
- M2

LLM CORE

Ship Project 1 (RAG); start post series.
- M3

AGENTS

Ship Project 2; publish blog #1-2.
- M4

DEPLOY

Fine-tune + eval; launch full-stack AI app.
- M5

PORTFOLIO

4 projects live; 1 OSS PR; mock interviews.
- M6

LAND IT

Apply 5-10/wk; system design; negotiate.

START THIS WEEK Immediate Next Actions

- 📅 Calendar-block 10-12 focused hrs/week.
- 📅 Enroll in an ML course today.
- 📅 Set up GitHub + dev env + an LLM API key.
- 📄 Rewrite LinkedIn: "Web Dev + Data → AI Engineer (in progress)".
- 📄 Choose Project #1 idea + dataset this week.
- 📄 Post your "Day 1" commitment publicly for accountability.